Stable matchings

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Outline

Why Stable Matchings?

Mathematics and Real Life

Basic examples

Looking For a Stable Matching

Gale–Shapley Algorithm

Correctness Proof

Why the Algorithm is Unfair

Why The Algorithm is Very Unfair

University Entrance in Russia

Old system:

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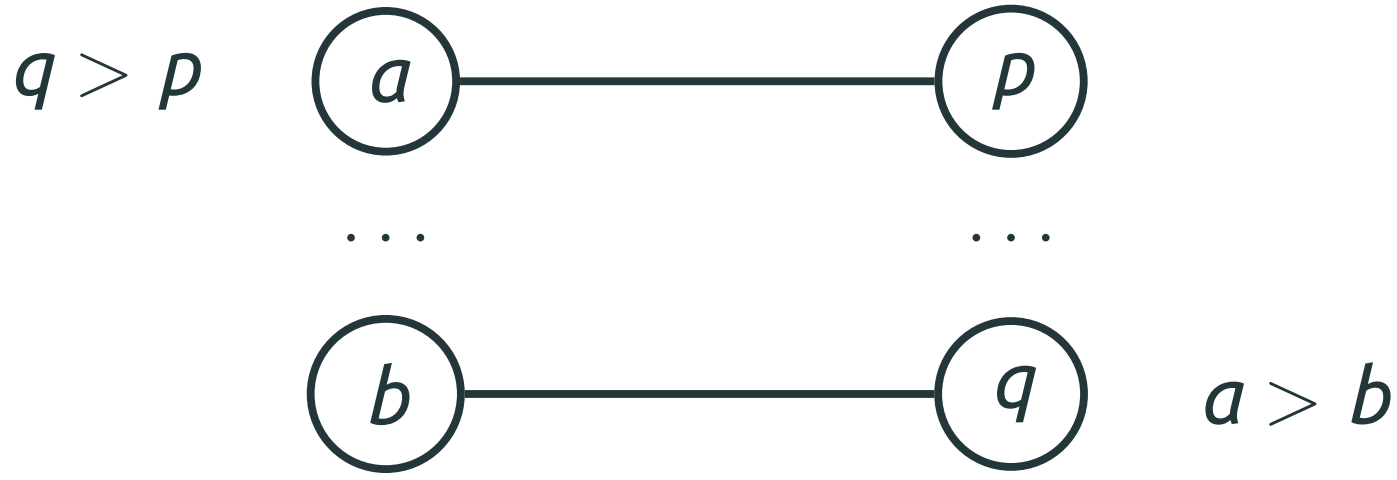
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- *n* positions
- full information: lists of preferences
- matching: *n* pairs (candidate, position)
- graph theory: “perfect matching”
- minimal requirement: stability

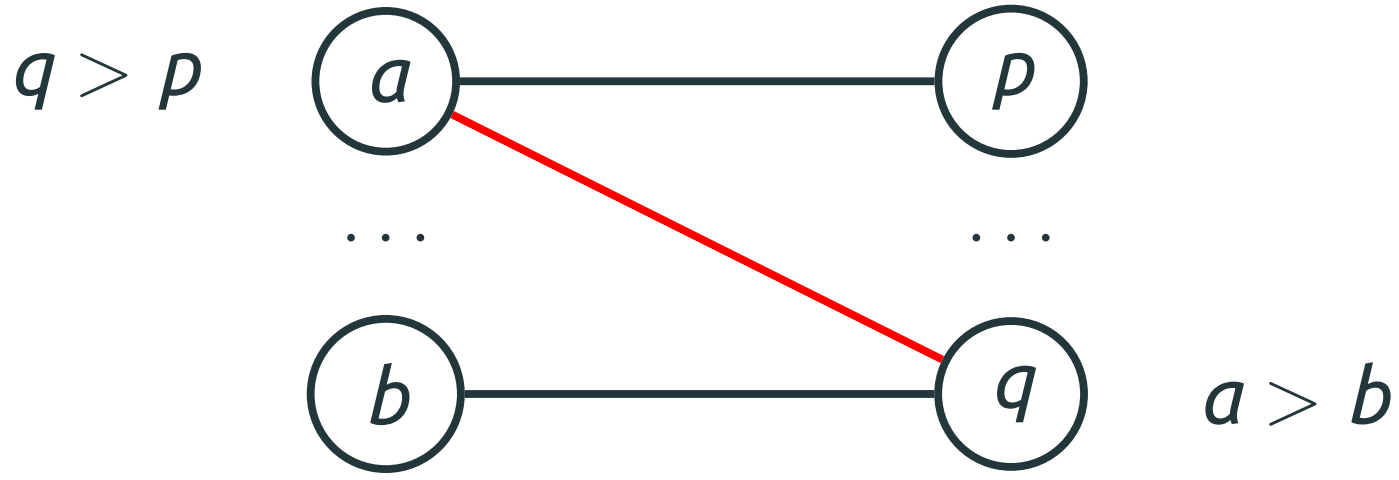
Unstable Matching



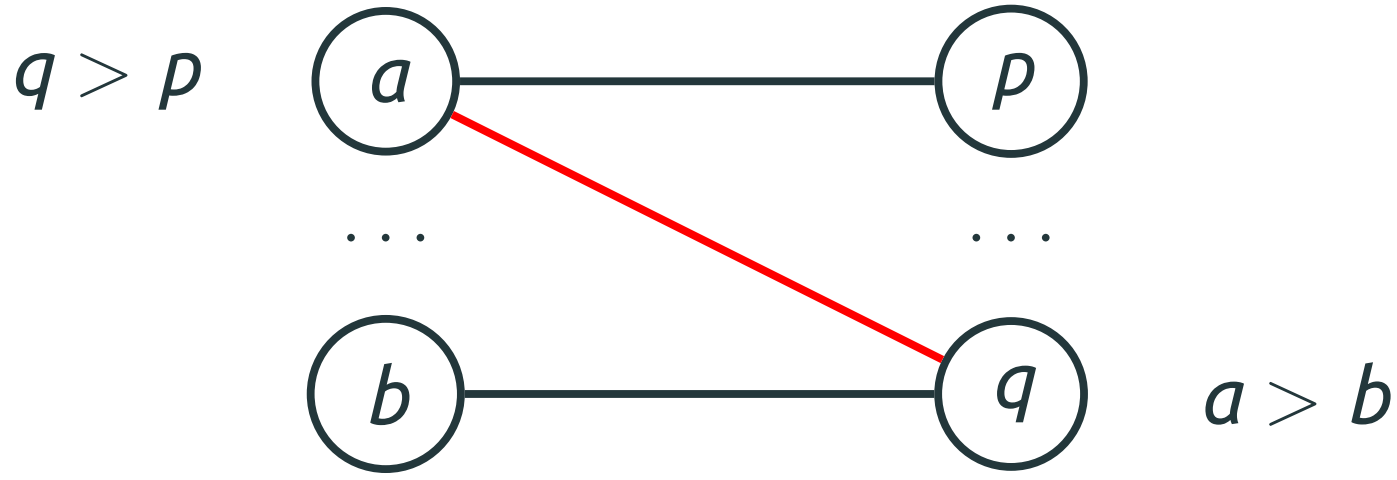
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Is there a stable matching?

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…by Writing to a Wide Audience

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- David Gale (1921–2008), Lloyd Shapley: College Admissions and the Stability of Marriage, 1962

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- American Mathematical Monthly
  `https://www.jstor.org/stable/2312726`

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Unique Stable Marriage

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- unique stable marriage: why?

The Same Ordering

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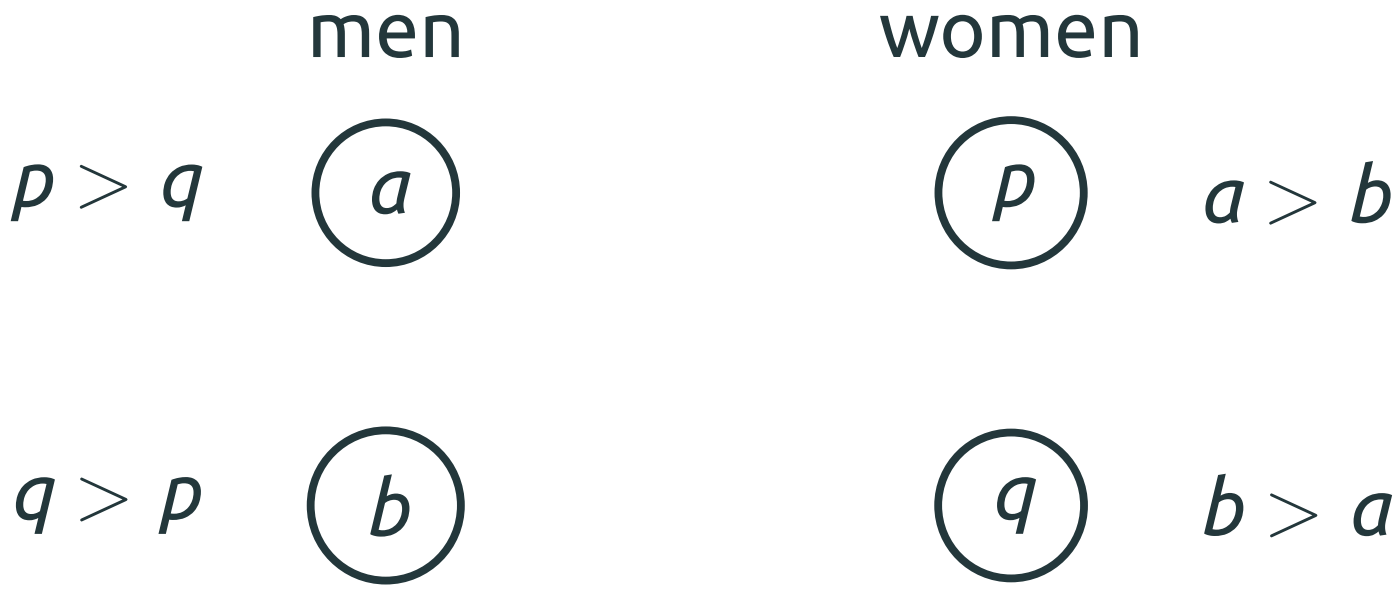
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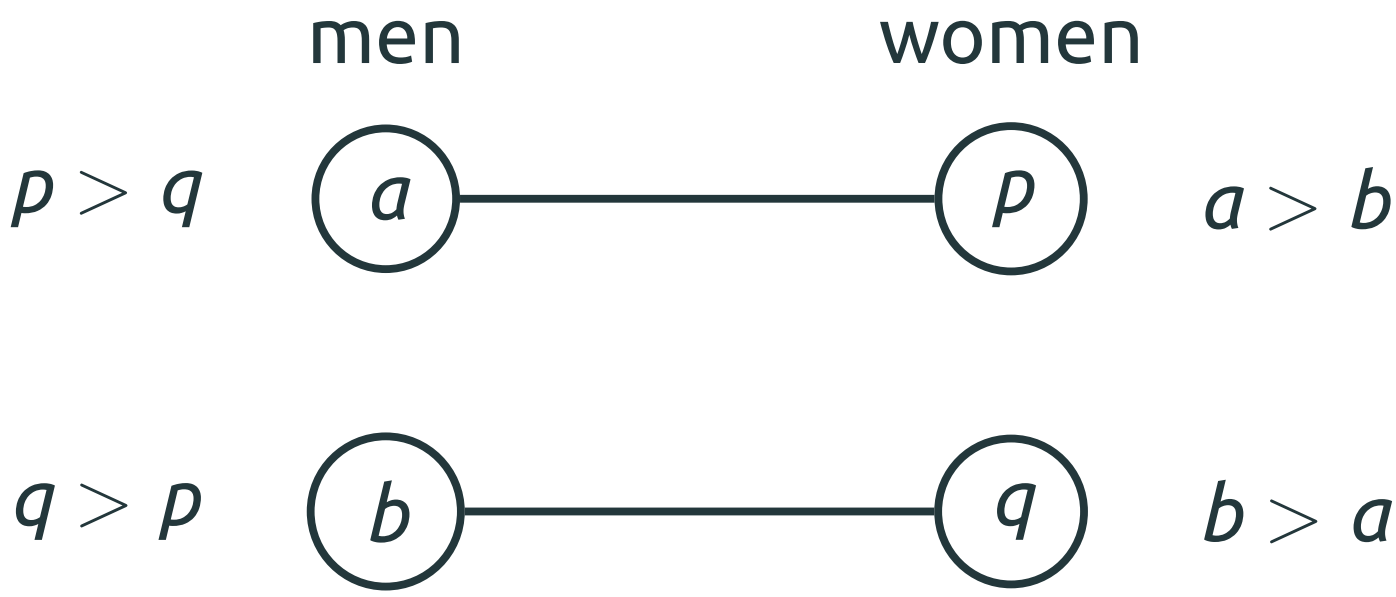
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- symmetry

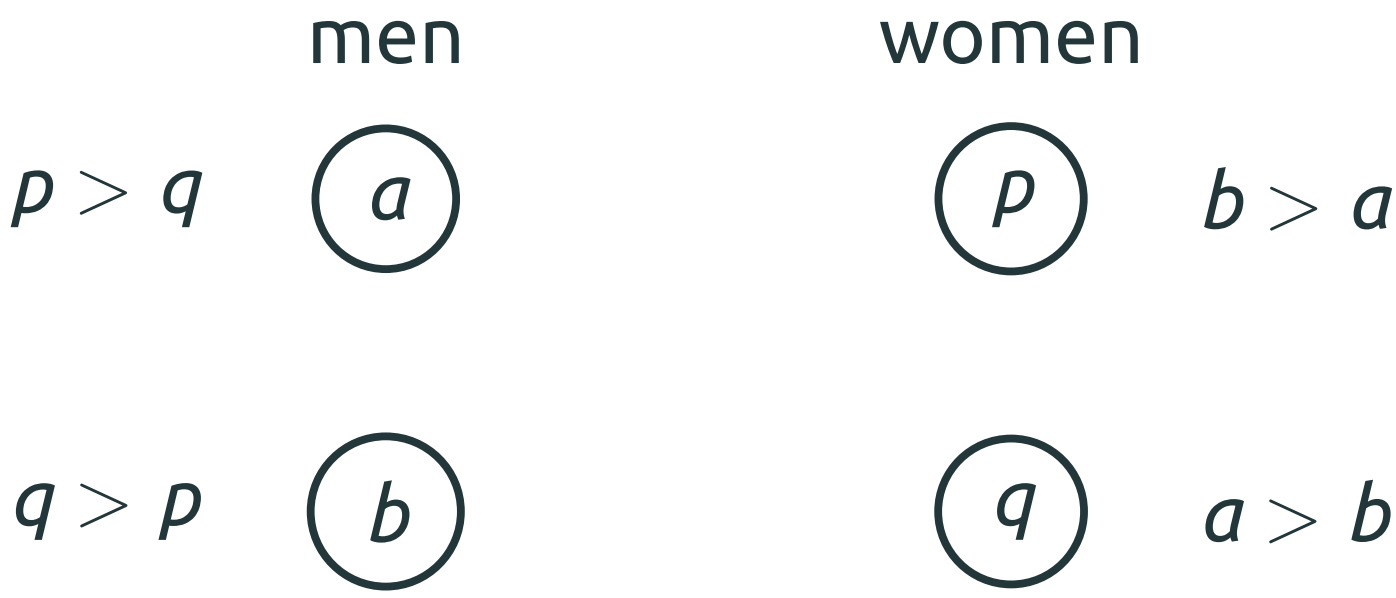
$n = 2$: **Everyone Is Happy**



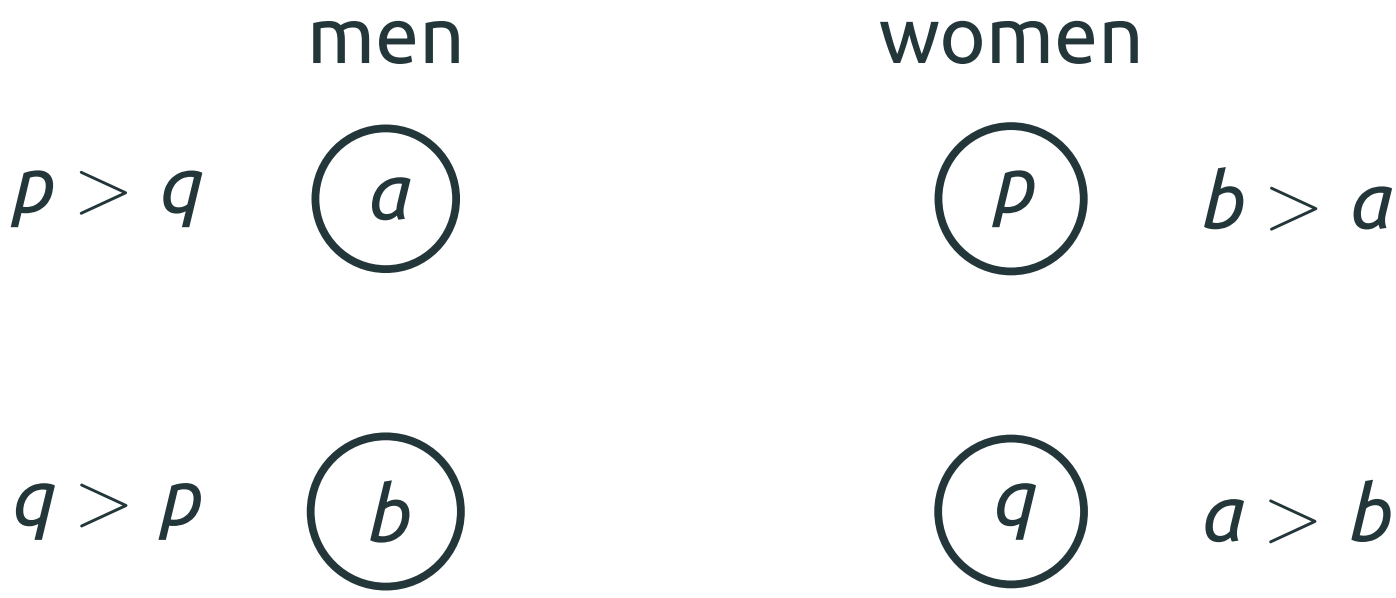
$n = 2$: Everyone Is Happy



$n = 2$: Two Stable Matchings

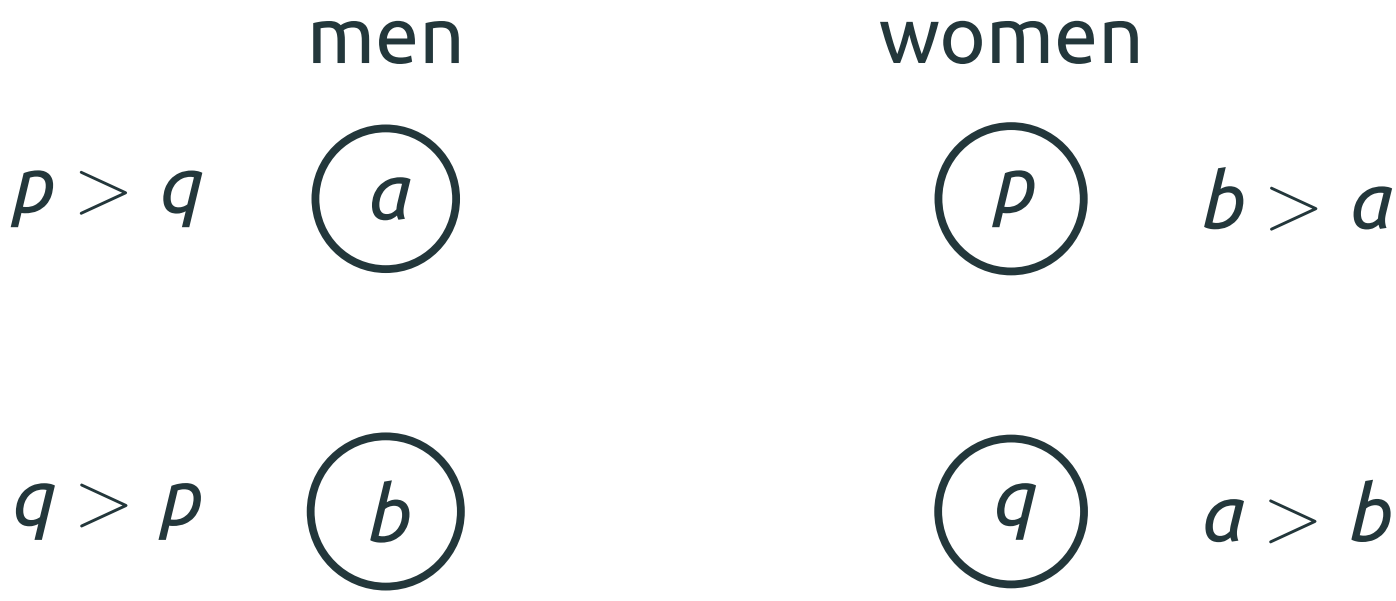


$n = 2$: Two Stable Matchings



two bad but stable matchings

$n = 2$: Two Stable Matchings



two bad but stable matchings

H. Heine / R. Schumann (“Dichterliebe”)

`https://www.youtube.com/watch?v=UzfWyCLmHLc`

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- repeat until a stable matching is obtained

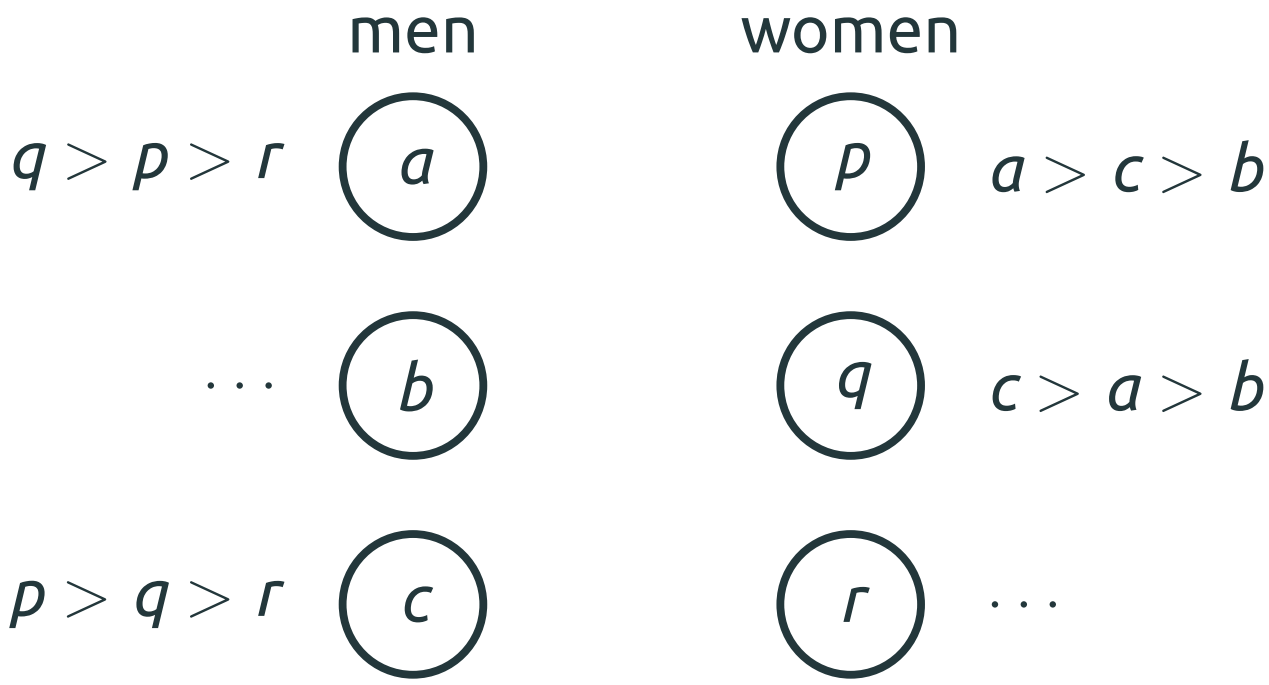
Stabilization?

Stabilization?

…not guaranteed

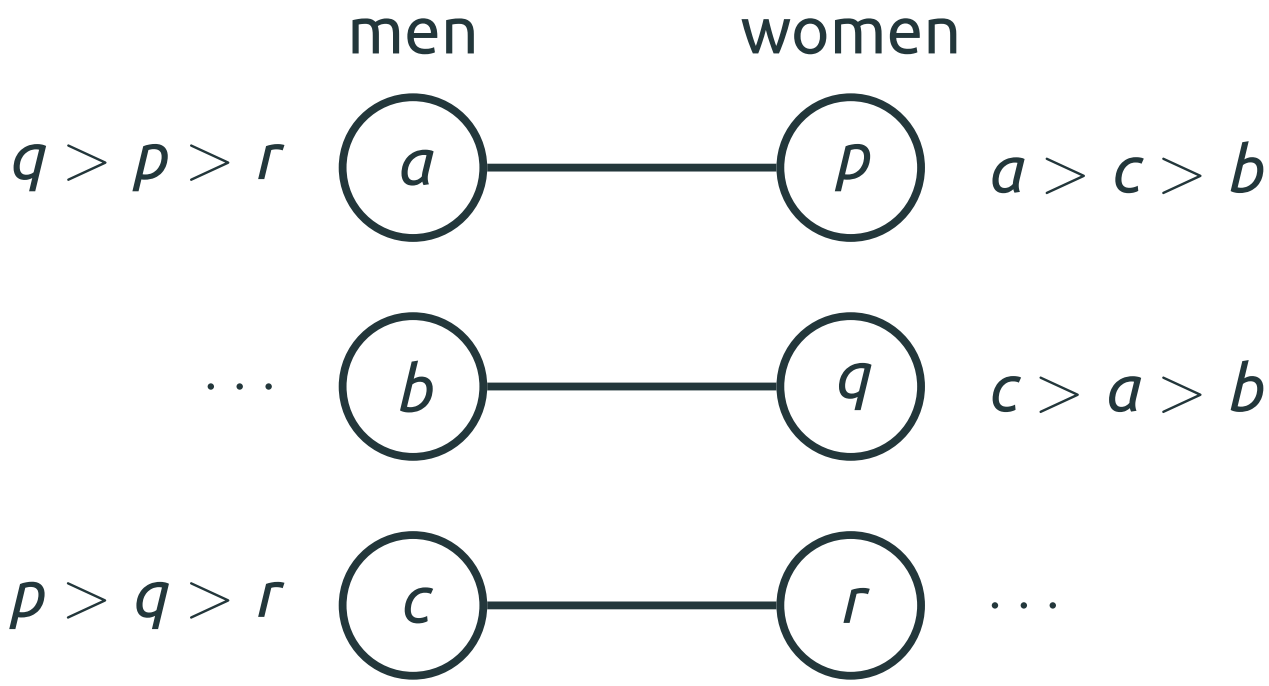
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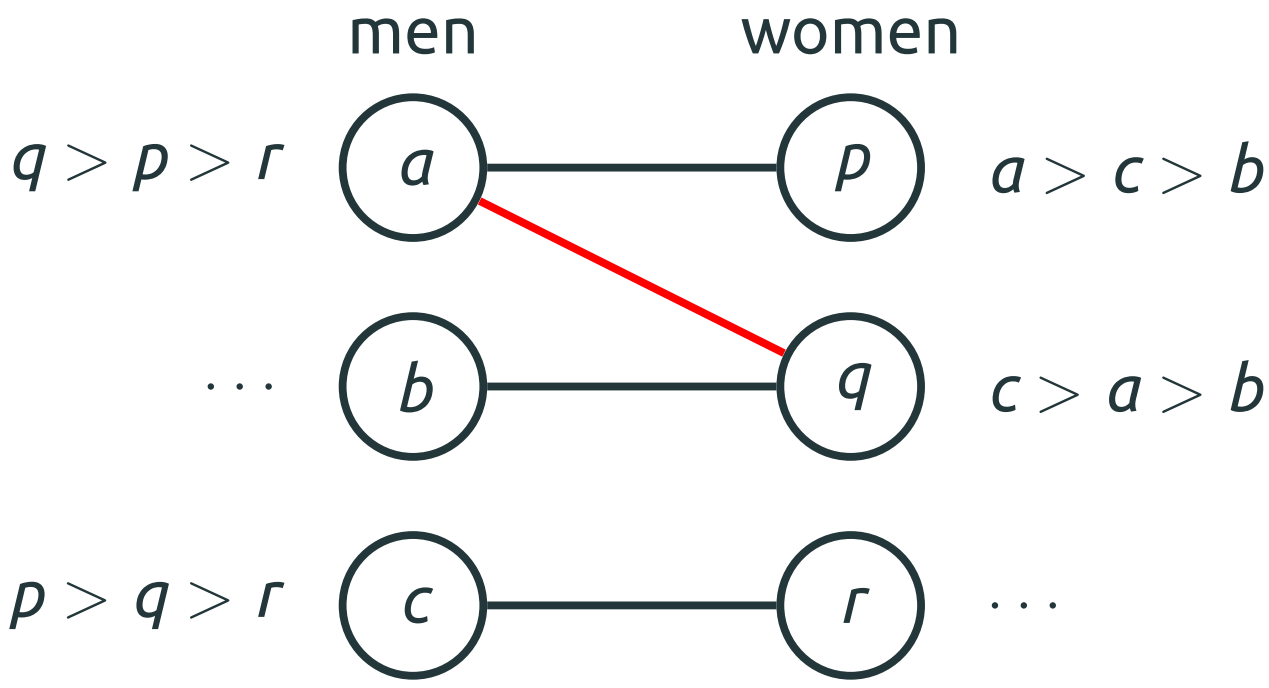
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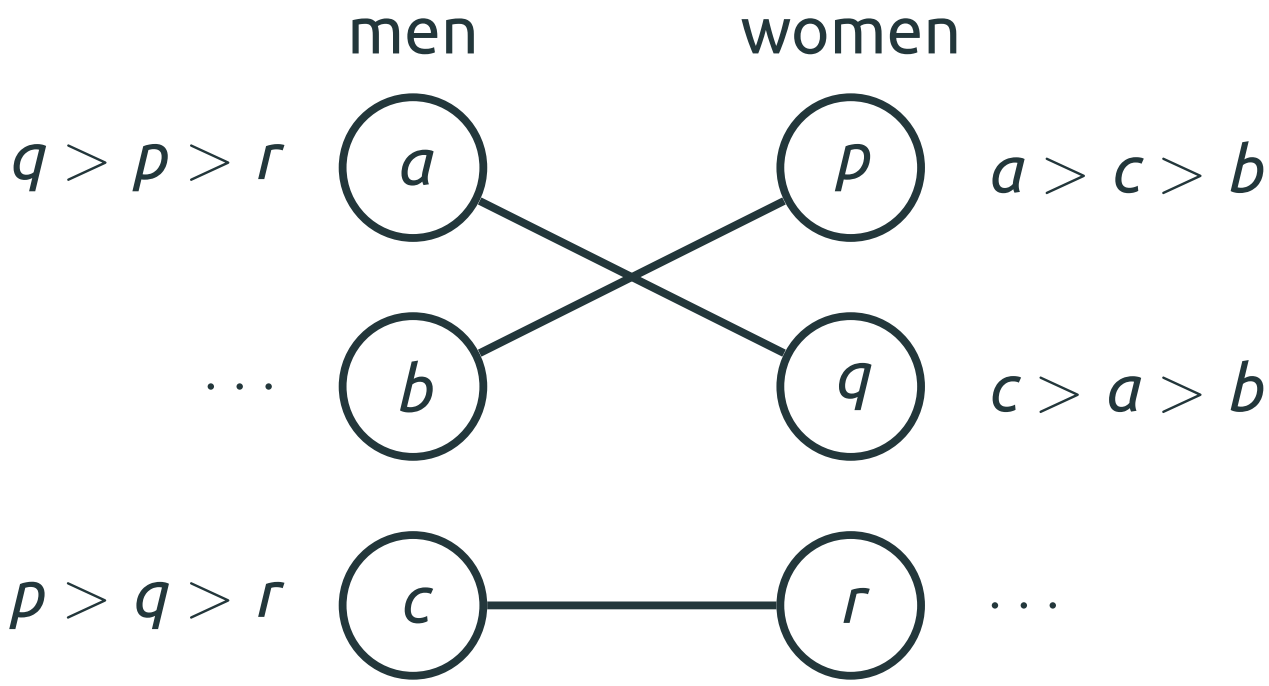
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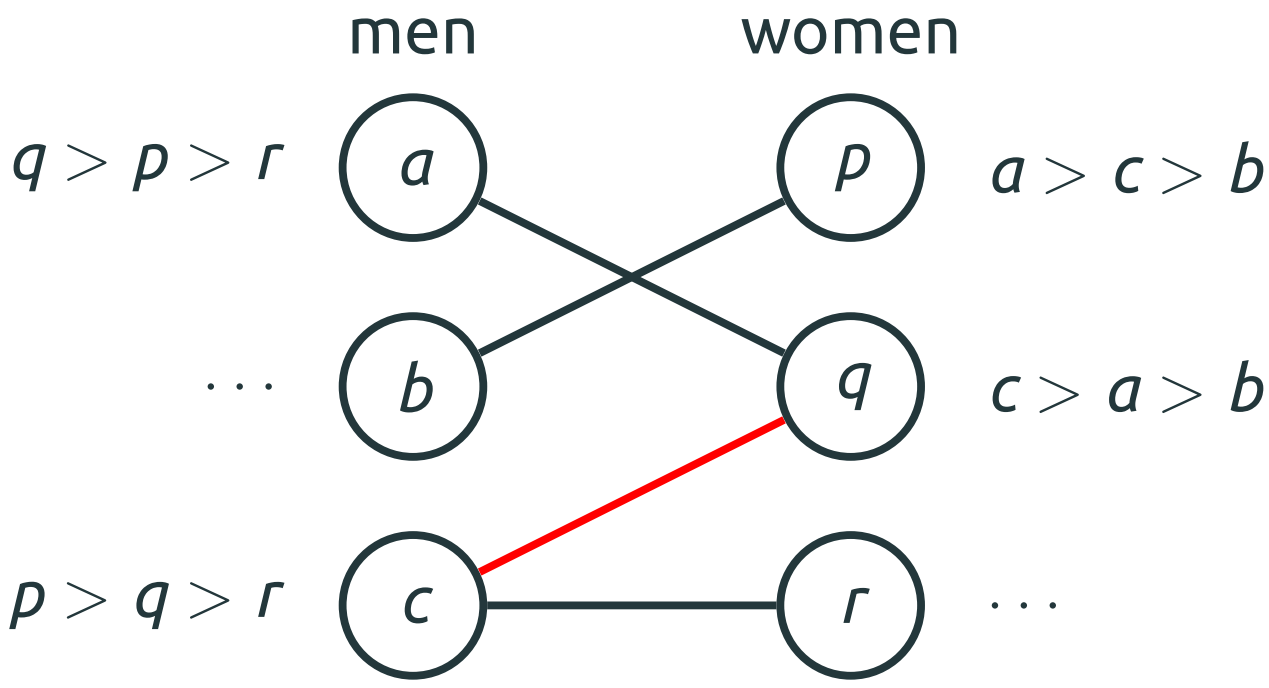
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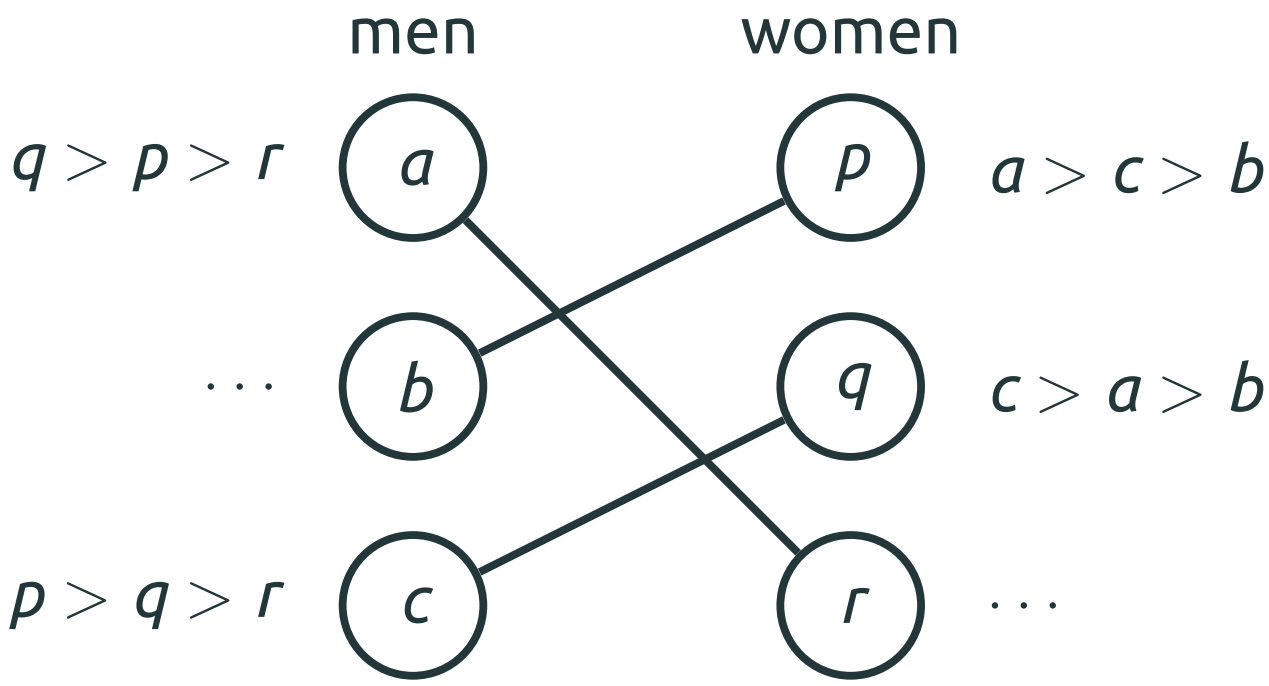
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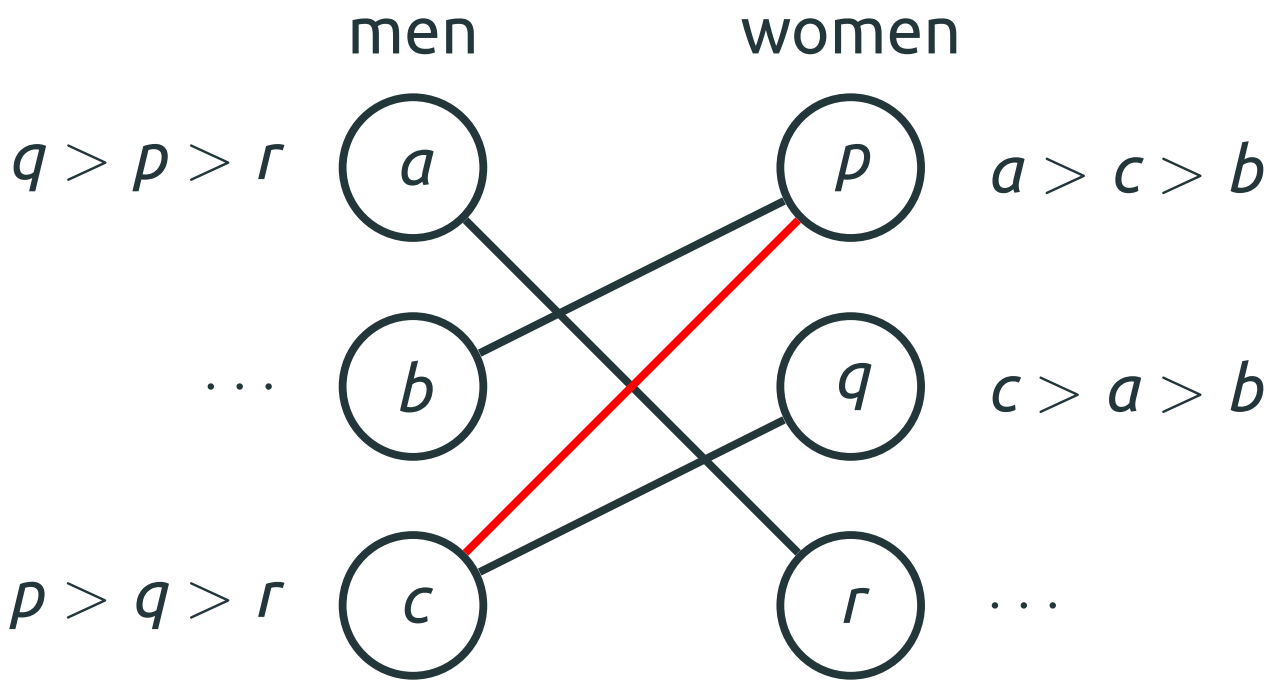
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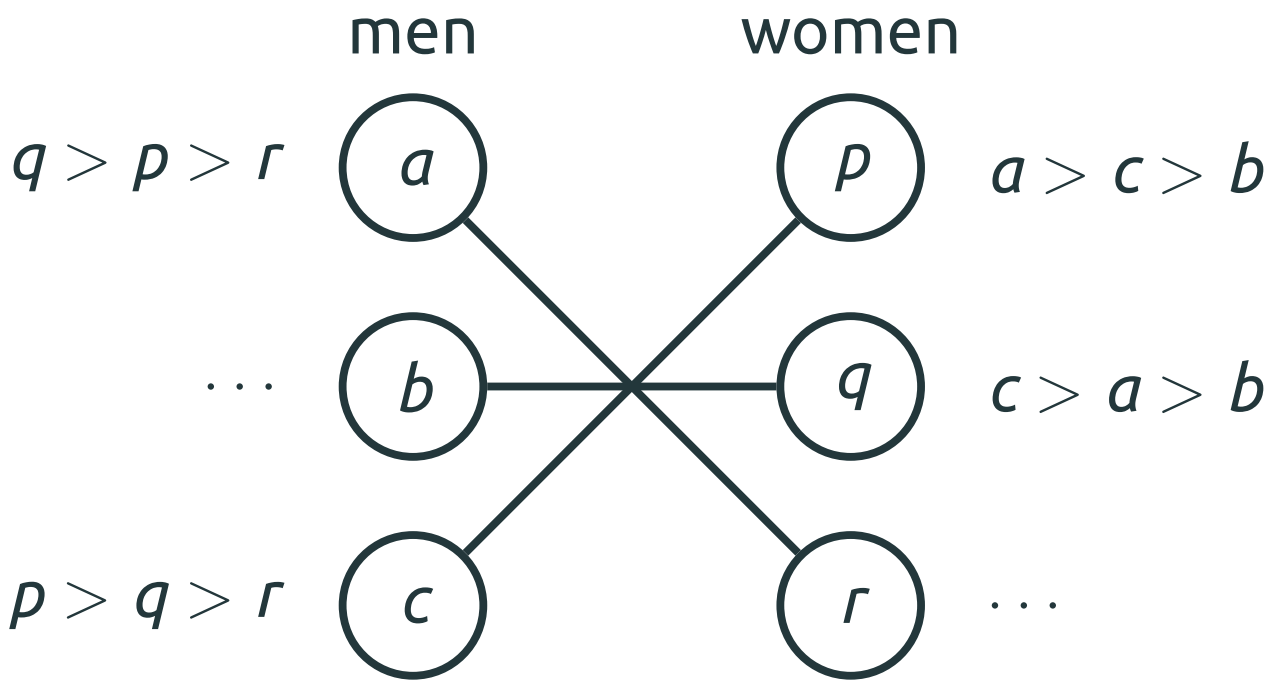
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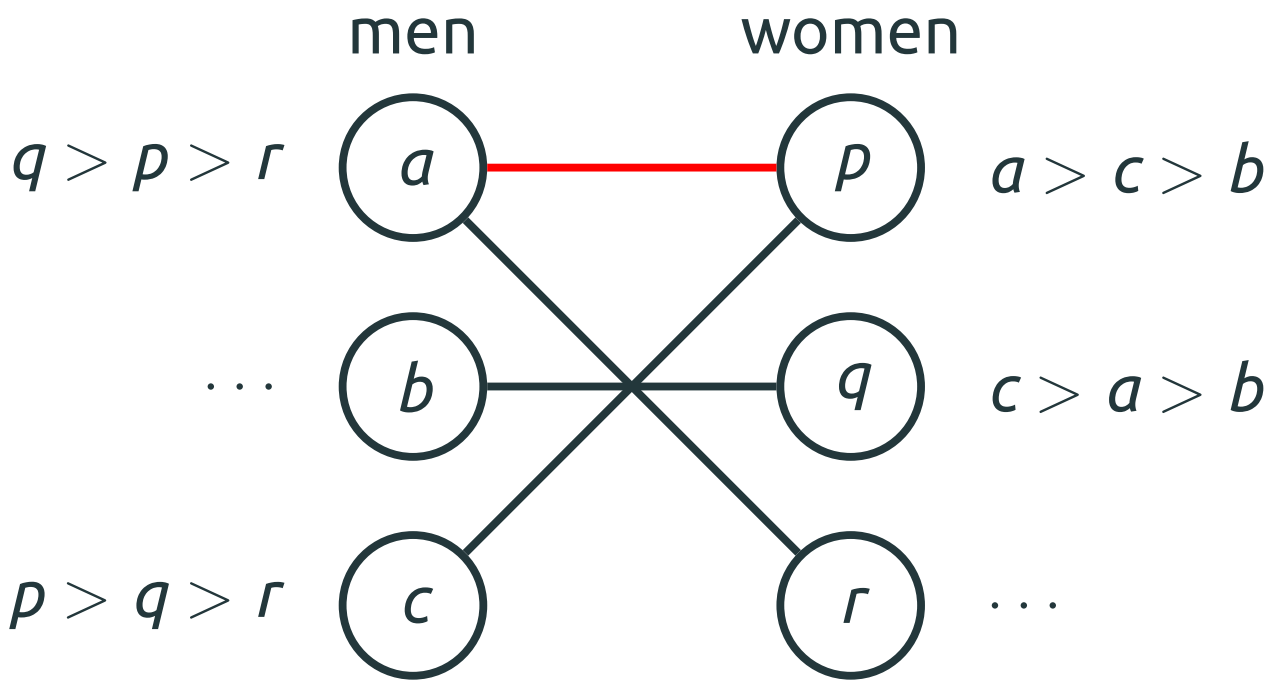
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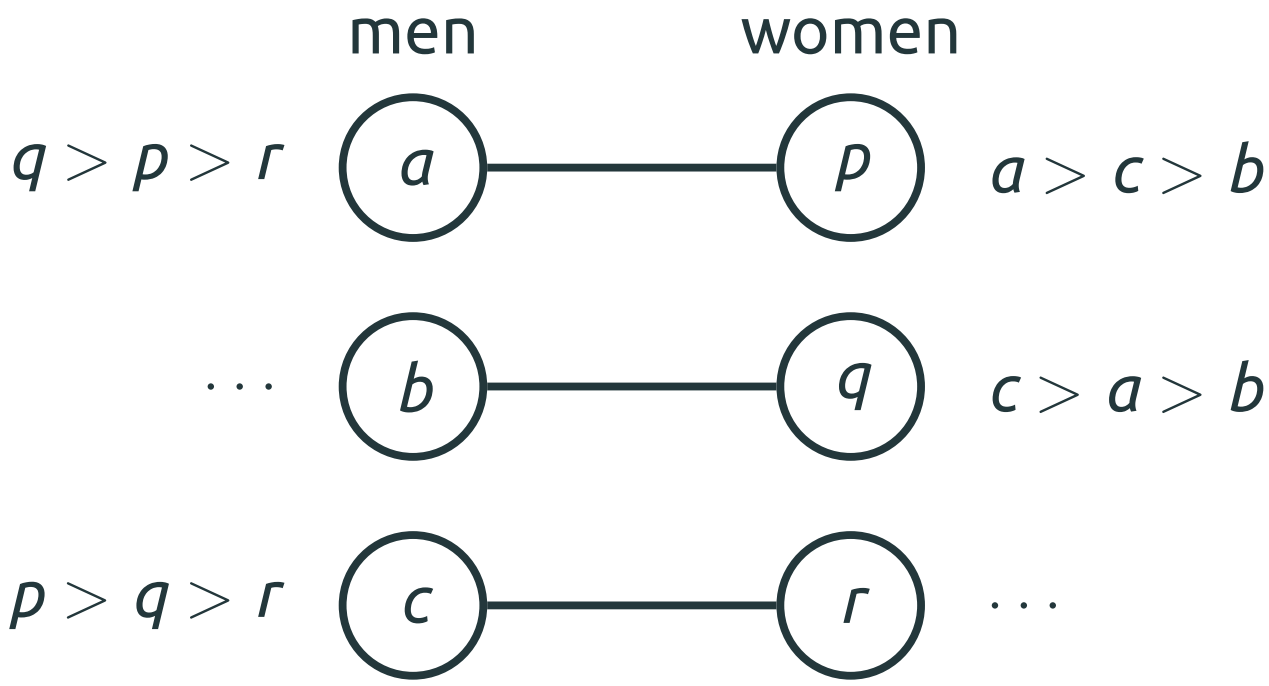
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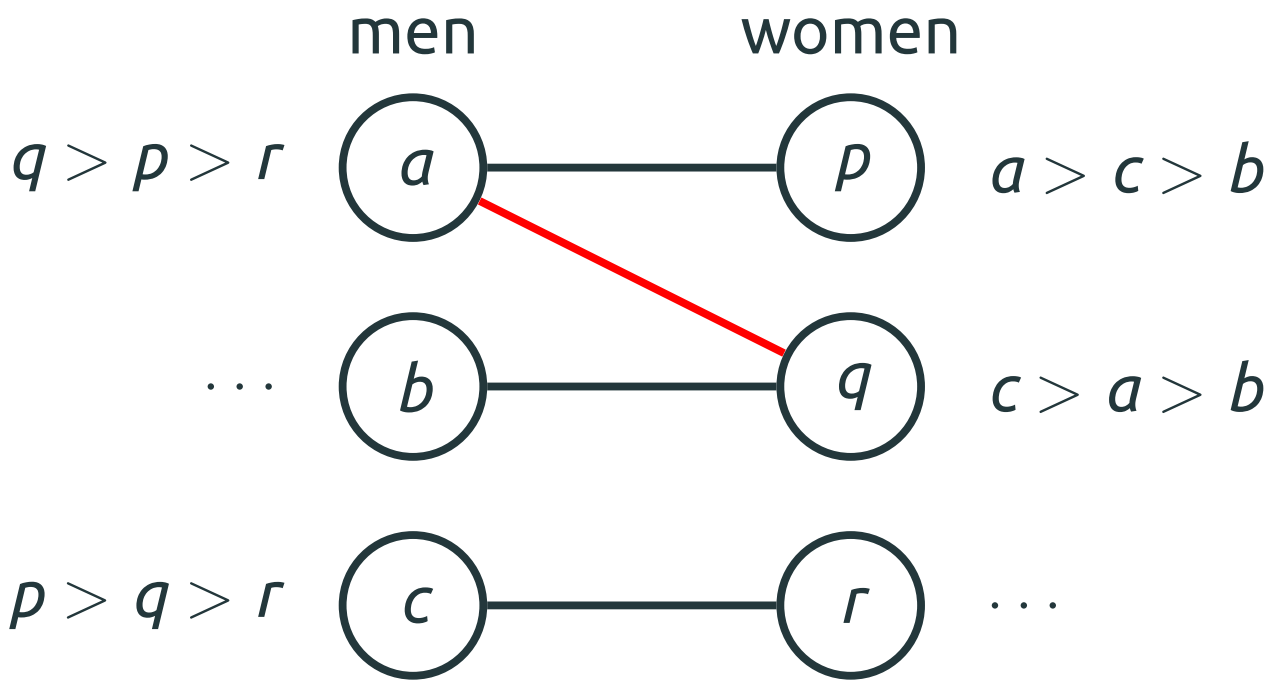
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- non-deterministic

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- stability

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- contradiction (#women = #men)

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- corollary: non-determinism inessential
- corollary: no collisions if for every man the best possible partner (in stable matchings) is chosen
- reformulation: if m was rejected (in any way) by w during the algorithm, (m, w) cannot appear in a stable matching

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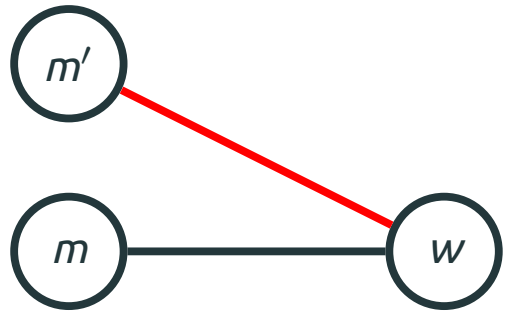
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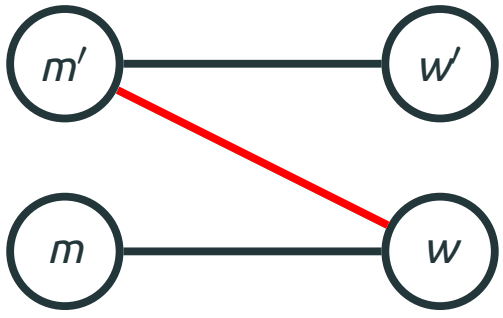
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- induction by the rejection moment
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- but w rejected/left m because $w : m' > m$



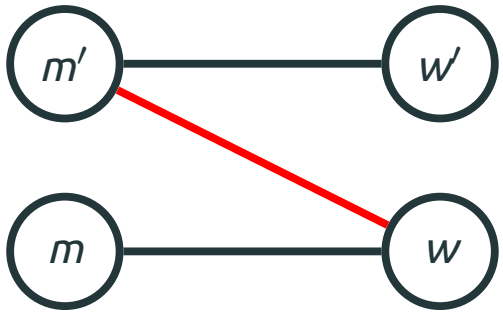
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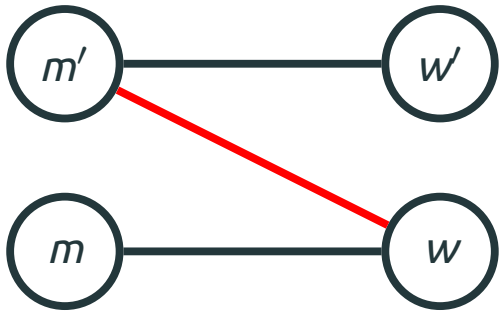
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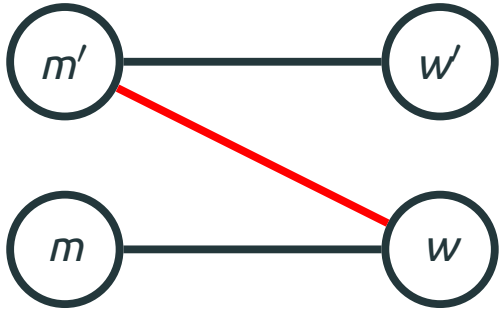
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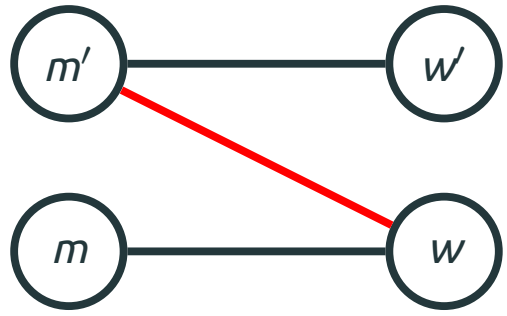
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Outline

Why Stable Matchings?

Mathematics and Real Life

Basic examples

Looking For a Stable Matching

Gale–Shapley Algorithm

Correctness Proof

Why the Algorithm is Unfair

Why The Algorithm is Very Unfair

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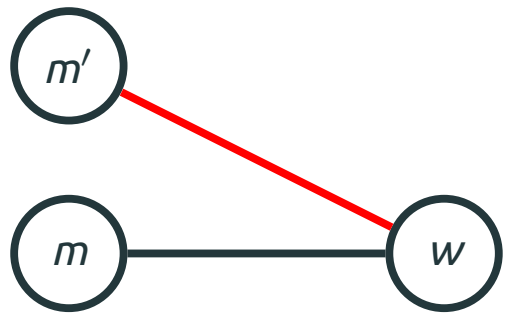
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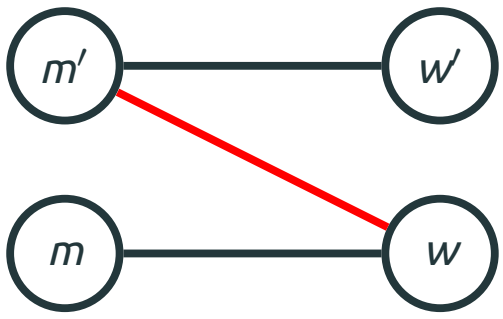
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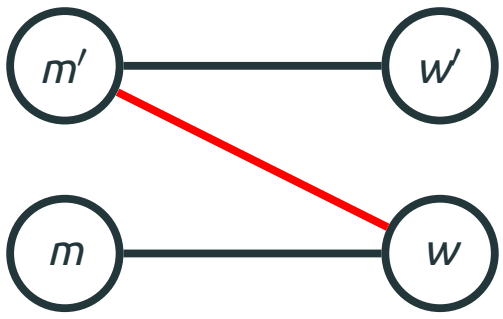
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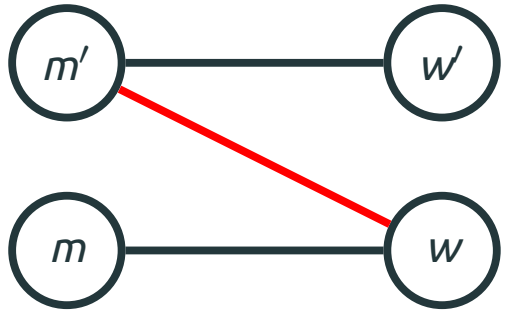
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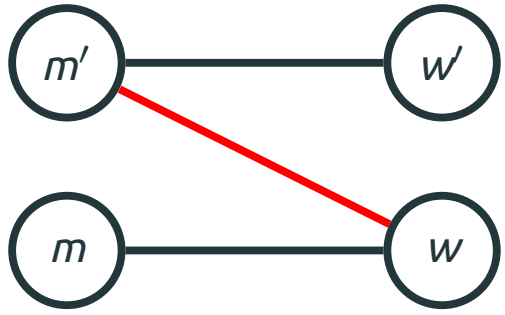
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